TEXORA · LECTURE SHEET

Fancy Designs of Fabrics

Beyond the fundamentals lie the fancy structures: weaves combined in one cloth, orders rearranged until the surface turns granular (crepe), cells sunk and ridged by organised long floats (honeycomb), openwork imitated without doups (mock leno), and cords built from paired systems (Bedford cord). Each is a construction method, not a mystery — and each design in this sheet has been drawn and float-checked square by square.

TE-3205 · FABRIC STRUCTURE AND DESIGN · TOPIC 5 · LEVEL 3 TERM 2

1. How fancy designs are made — the four methods

- **Combination:** two or more weaves placed side by side (stripes, checks) or merged into one repeat (Topic 3, Section 5).
- **Rearrangement:** the ends or picks of a base weave are taken in a new order, breaking its lines (crepe, Section 3).
- **Added interlacing / float organisation:** extra marks inserted into a ground, or floats deliberately concentrated to build relief (honeycomb, Section 4).
- **Extra systems:** additional warp or weft threads that figure, cord or pile on a ground weave (Bedford cord, Section 6; pile, Topic 8; extra-weft figuring in brocades).

2. Combined stripe designs

The simplest fancy construction: bands of different weaves side by side in the warp direction. The repeat in ends is the sum of the band widths; the pick repeat is the LCM of the bands' pick repeats. Figure 1 combines 6 ends of plain weave with 6 ends of 2/2 twill: pick repeat = LCM(2, 4) = 4.

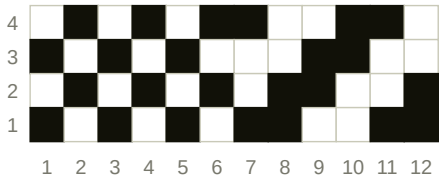


Figure 1 — Combined stripe: 6 ends plain + 6 ends 2/2 twill, repeat 12 ends × 4 picks. Verify the join: at the boundary (end 6 | end 7) and at the repeat edge (end 12 | end 1 of the next repeat) no float exceeds the parent weaves' — a join that creates a long float is the classic defect of stripe design and must be checked exactly here.

Weave checks are the same idea in two directions (bands by ends and by picks). Practical rule: adjoining weaves should have compatible **contraction** — a plain-weave band crimps and contracts far more than a satin band beside it, and the cloth cockles unless setts or yarn tensions compensate. This, not drawing, is what makes stripe design a mill skill.

3. Crepe weaves

A crepe fabric shows a fine, random, granular (seersucker-like in miniature, "orange-peel") surface with **no repeat visible to the eye**, although it is woven from a small exact repeat. The effect is produced by destroying every line: binding points are scattered irregularly while floats are kept short. Standard constructions:

- **Rearranged base weave:** a satin or twill is redrawn with its ends in a broken order (e.g. the satin order 1, 4, 7, 2, 5, 8, 3, 6 applied to a plain-ground design).
- **Ground + added marks (used for Figure 2):** start from plain weave and **add one extra warp-up mark per pick in satin order** — here step 3 starting from end 2, putting the added marks at ends 2, 5, 8, 3, 6, 1, 4, 7 on picks 1–8. Each added mark lengthens some floats and breaks the plain alternation at an irregular place.
- **Rotation / reversal of a small motif:** a small weave unit is repeated in four rotations to cancel its direction.

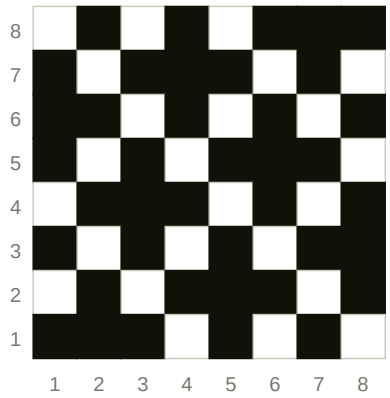


Figure 2 — Crepe weave, 8 × 8: plain weave ground (compare Figure 1 of Topic 1) with one added mark per pick in satin order (step 3 from end 2). Float-checked: the longest float anywhere in the repeat is **3**, no diagonal or straight line survives, and every row and column interlaces irregularly — the three tests a crepe design must pass.

Crepe fabrics (true woven crepes, distinct from crepe *yarn* effects made with high-twist yarns on plain weave) are used for dress fabrics where a lively, non-shiny, crease-hiding surface is wanted. Weaving note: because adjacent ends lift irregularly, crepe designs need enough shafts for the full repeat — here 8.

4. Honeycomb weave

Honeycomb builds a **cellular surface**: square or diamond cells whose centres are recessed and whose edges stand as ridges. The mechanism is float organisation: in the **centre** of the cell both warp and weft run in long floats on the back, so in finishing those threads contract and pull the centre in; along the **cell borders** the threads interlace in a compact diamond of warp-up marks and the neighbouring cells' quarter-blocks, forming the firm ridge. Figure 3 shows one full repeat of the ordinary 8 × 8 honeycomb.

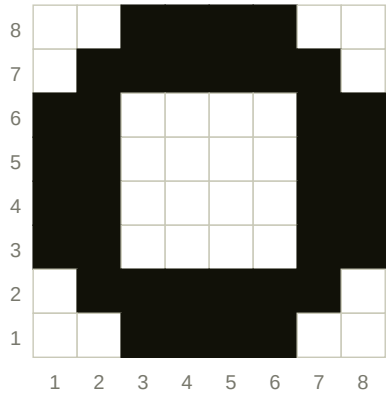


Figure 3 — Honeycomb (waffle) weave, repeat 8 × 8. Read it as: a warp-up diamond ring and the corner blocks of the four neighbouring cells stand as ridges; the central 4 × 4 area is left weft-up (warp down) — its long back floats contract in finishing and sink the cell. Float-checked: the ridge threads float **6** (the defining long float of the structure, along the second row and column of the repeat), the centre floats are 4, and the design is symmetric under both mirror flips, as a cell design must be.

Because the ridge floats are long, honeycomb is woven in yarns that tolerate it (cotton, linen) and used where the cells themselves are the function: **honeycomb towels and bath mats** (the cells hold water and the fabric dries fast), cellular blankets and dress fabrics. Larger repeats (10 × 10, 12 × 12) deepen the cells and lengthen the ridge floats further.

5. Mock leno

True leno (doup) weaving crosses pairs of ends around each pick to lock open meshes; **mock leno** imitates the open, locked look with ordinary shafts. Warp ends are dented in **groups (typically 3 ends, then a dent left empty)**, and the design makes the ends of each group interlace in plain order but in opposite phase to the neighbouring group, with the group spreading against the empty dent. The result is a grid of small locked apertures — curtain and dress fabrics with an open texture that does not slip, because the opposed plain interlacings pinch the picks between groups. The design rule to verify: within each group the interlacing is plain; the aperture is a *denting and grouping* effect — the design grid alone, without the denting plan, does not produce it (recall Topic 1: denting is a full record, not a footnote).

6. Bedford cord

Bedford cord raises **rounded cords running warp-way**, separated by fine sunken lines. Construction: the warp is arranged in groups; within each cord the ends weave **plain weave with a normal (cord) weft**, while one or two **cutting ends** between cords weave so as to pull the cord edges down (they bind tightly, typically plain in opposition), forming the sunken line. Alternate picks may be **wadding picks** — thick, soft weft inserted under the cord only (floating on the back between cords) — to make the cord fuller and rounder without stiffening the cloth. The design is drawn as columns: cord columns in plain order, cutting columns in opposed order, and the wadding picks marked to pass behind the cords. Uses: trousers and uniform fabrics ("Bedford cord" trousering), upholstery. The examination point is the division of labour: plain interlacing makes the cord body, the cutting ends make the groove, the optional wadding pick makes the roundness.

7. Extra figuring — the bridge to jacquard

When pattern threads are added only where the figure needs them — an **extra weft** floated across the back between motifs, or an **extra warp** figuring over a plain ground — the design is built by superimposing the figure weave on the ground weave's design, square by square: ground marks first, figure marks replacing them where the motif demands, with the float lengths of the extra system checked and (for weft figuring) the waste weft afterwards clipped or left as backing. This superposition discipline is the whole of figured design in miniature, and it is also how the satin-stripe and double cloths of Topics 4 and 7 are drawn.

Glossary

Crepe weave — small-repeat weave with scattered binding points and short floats, giving a granular surface. **Honeycomb** — cellular weave; recessed centres with floating threads, ridged interlaced borders. **Mock leno** — openwork imitation of leno by grouped denting and opposed plain interlacing. **Bedford cord** — warp-way cords in plain weave with cutting ends and optional wadding picks. **Wadding pick** — thick pick floating behind a cord or cloth layer to add fullness.

Key points

• Fancy = combination, rearrangement, float organisation, or extra systems — name the method before drawing.

• Crepe: scatter the binding, keep floats ≤ ~3, kill every line. Honeycomb: sink the centre with floats (centre 4), stand the ridge (float 6), in an 8 × 8 cell.

• Mock leno lives in the denting plan; Bedford cord divides work between cord ends, cutting ends and wadding picks.

Practice

1. A stripe design combines 8 ends of plain weave with 8 ends of 3/1 twill. State the end repeat and the pick repeat.
2. In Figure 2, find the added mark in pick 3 and state the float it creates for end 8, verifying the "no float over 3" claim for that end.
3. Why does a honeycomb fabric sink its cells only after weaving/finishing, although the design already shows the cell pattern on paper?
4. Mock leno and true leno can look alike. State the structural difference that a buyer could test by pulling one pick.

Answer key

1. Ends: 8 + 8 = **16**. Picks: LCM(2, 4) = **4**.
2. Pick 3's added mark is at end 8 (order 2, 5, 8, …). In the plain ground end 8 is up on picks 2, 4, 6, 8; the added mark fills pick 3, joining picks 2–4 into a single run of **3** up-picks. Its remaining runs are single picks in both states (up on 6 and 8, down on 1, 5, 7, and pick 8 does not join pick 1 across the boundary since end 8 is down on pick 1) — so the longest float of end 8 is exactly 3, as claimed.
3. On paper the "cell" is only a float plan. The sinking is mechanical: the long back floats of both systems in the cell centre are under no interlacing restraint, so when the cloth relaxes/contracts in finishing they draw together and pull the centre down, while the frequently interlaced borders resist contraction and stay up as ridges.
4. In true leno the ends of each pair physically cross around the pick (doup crossing), so a pick cannot be withdrawn without uncrossing them; in mock leno no end crosses any other — the pick is held only by the opposed plain interlacings and the group spacing, and can be drawn out endwise like a plain-weave pick, leaving the groups standing.